

Data Fundamentals for BI in a Business

Agenda

- Understanding data in the context of business operations
- Data types, sources, and quality
- Data governance and compliance considerations for business

Data Quality: Why Preprocess the Data?

- Measures for data quality (checking if information is good or bad):

A multidimensional view

- **Accuracy:** correct or wrong, accurate or not
- **Completeness:** not recorded, unavailable, ...
- **Consistency:** some modified but some not, dangling,..
- **Timeliness:** timely update?
- **Believability:** how trustable the data are correct?
- **Interpretability:** how easily the data can be understood?

Major Tasks in Data Preprocessing

- **Data cleaning**

- Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies

- **Data integration**

- Integration of multiple databases, data cubes, or files

- **Data reduction**

- Dimensionality reduction
- Numerosity reduction
- Data compression

- **Data transformation and data discretization**

- Normalization
- Concept hierarchy generation

Data Cleaning

- Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
 - **incomplete**: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., *Occupation*=" " (missing data)
 - **noisy**: containing noise, errors, or outliers
 - e.g., *Salary*="−10" (an error)
 - **inconsistent**: containing discrepancies in codes or names, e.g.,
 - *Age*="42", *Birthday*="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
 - **Intentional** (e.g., *disguised missing* data)
 - Jan. 1 as everyone's birthday?

Data Cleaning: Incomplete (Missing) Data

- **Missing data may be due to**
 - equipment malfunction
 - inconsistent with other recorded data and thus deleted
 - data not entered due to misunderstanding
 - certain data may not be considered important at the time of entry
 - not register history or changes of the data
- Missing data may need to be inferred

How to Handle Missing Data?

- **Ignore the tuple:** deleting the entire record (row) if it has missing data — not effective when the % of missing values per attribute varies considerably
- **Fill in the missing value manually:** looking at each missing value and trying to find the correct information. This is accurate but very time-consuming and often impossible if you have a lot of data.
- **Fill in it automatically with**
 - a global constant : e.g., “unknown”, a new class?!
 - the attribute mean (average)
 - the attribute mean for all samples belonging **to the same class** (smarter)
 - **the most probable value:** Try to predict the most likely value based on other information by using inference-based such as Bayesian formula or decision tree

Data Cleaning: Noisy Data

- **Noise:** random error or variance in a measured variable. It makes the data less clear and harder to analyze
- **Incorrect attribute values may be due to**
 - faulty data collection instruments
 - data entry problems
 - data transmission problems
 - technology limitation
 - inconsistency in naming convention

How to Handle Noisy Data?

■ Binning

- first **sort** data and **partition** into (equal-frequency) **bins**
- **Example:** ages: 10, 12, 15, 18, 20, 22, 25, 28, 30. You could create **three bins: (10-19) (20-29) (30+)**.
- **Smoothing by bin means:** Replace each age in the 10-19 bin with the average age of that bin (which would be around 14). Do the same for the other bins.
- **Smoothing by bin median:** Replace each age in the 10-19 bin with the middle age of that bin.
- **Smoothing by bin boundaries:** Replace each age in the 10-19 bin with the closest bin boundary (either 10 or 19).

How to Handle Noisy Data?

- **Regression**

- smooth by fitting the data into regression functions

- **Clustering**

- Groups similar data points together
- detect and remove outliers (are often far away from the clusters)

- **Combined computer and human inspection**

- detect suspicious values and check by human (e.g., deal with possible outliers)

Data Cleaning as a Process

- It's like detective work to find errors and then fixing them.
- **Data discrepancy detection:** finding data that doesn't make sense
 - Use metadata (e.g., domain, range, dependency, distribution)
 - **Check field overloading:** one field is used to store multiple pieces of information. (A "contact" field might contain both a phone number and an email address)
 - **Check uniqueness rule** (Each customer ID should be unique), **consecutive rule** (numbers should usually be in sequence) and **null rule** (Certain fields, like "required fields", should not be empty)
 - **Use commercial tools designed to find data discrepancies**
 - **Data scrubbing:** This uses simple knowledge to find and fix errors. use simple domain knowledge (e.g., postal code, spell-check)
 - **Data auditing:** by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

Data Cleaning as a Process

- **Data migration and integration:** moving and combining data from different sources.
 - **Data migration tools:** transfer data and make changes during the transformation (A tool might convert dates from one format to another during migration)
 - **ETL (Extraction/Transformation/Loading) tools:** allow users to specify transformations through a graphical user interface

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Data Integration

- **Data integration:** Combines data from multiple sources into a coherent store
- **Schema integration:** matching up the *structure* of the data from different sources. e.g., A.cust-id $\equiv$ B.cust-#
- **Entity identification problem:** if records from different sources refer to the same real-world entity
 - e.g., "Bill Clinton" in one database the same person as "William Clinton" in another
- **Detecting and resolving data value conflicts:** For the same real-world entity, attribute values from different sources are different.
 - E.g., One database might say a customer's age is 30, while another says it's 32)
 - **Possible reasons:** different representations, different scales, e.g., metric vs. British units

Handling Redundancy in Data Integration

- Redundant data occur often when integration of multiple databases
 - ***Object identification:*** The same attribute or object may have different names in different databases
 - ***Derivable data:*** One attribute may be calculated (derived) from other attribute in another database, e.g., annual revenue
- Redundant attributes may be able to be detected by ***correlation analysis*** (how strongly two things are related) and ***covariance analysis*** (how two things change together)
- Careful integration of the data from multiple sources **improves mining speed and quality**

Handling Redundancy: Correlation Analysis (Nominal Data)

- **X² (chi-square) test:** is a statistical test that helps us understand if there's a relationship between two categorical variables (variables that can be divided into categories, like colors, types of fruit, or survey responses).

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

- The larger the X² value, the more likely the variables are related (**Strong Relationship**)
- The cells that **contribute** the most to the X² value are those whose actual count is very different from the expected count
- Correlation **does not imply causality** (Just because two things are related (correlated), it doesn't mean one causes the other.)
 - # of hospitals and # of car-theft in a city are correlated
 - Both are causally linked to the third variable: population

Chi-Square Calculation: An Example

	Play chess	Not play chess	Sum (row)
Like science fiction	250(90)	200(360)	450
Not like science fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

- Numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

- It shows that like_science_fiction and play_chess are correlated in the group

Handling Redundancy: Correlation Analysis (Numeric Data)

- Correlation coefficient (also called **Pearson's product moment coefficient**), is a statistical measure that expresses the extent to which two variables are linearly related, meaning how much they change together at a constant rate.

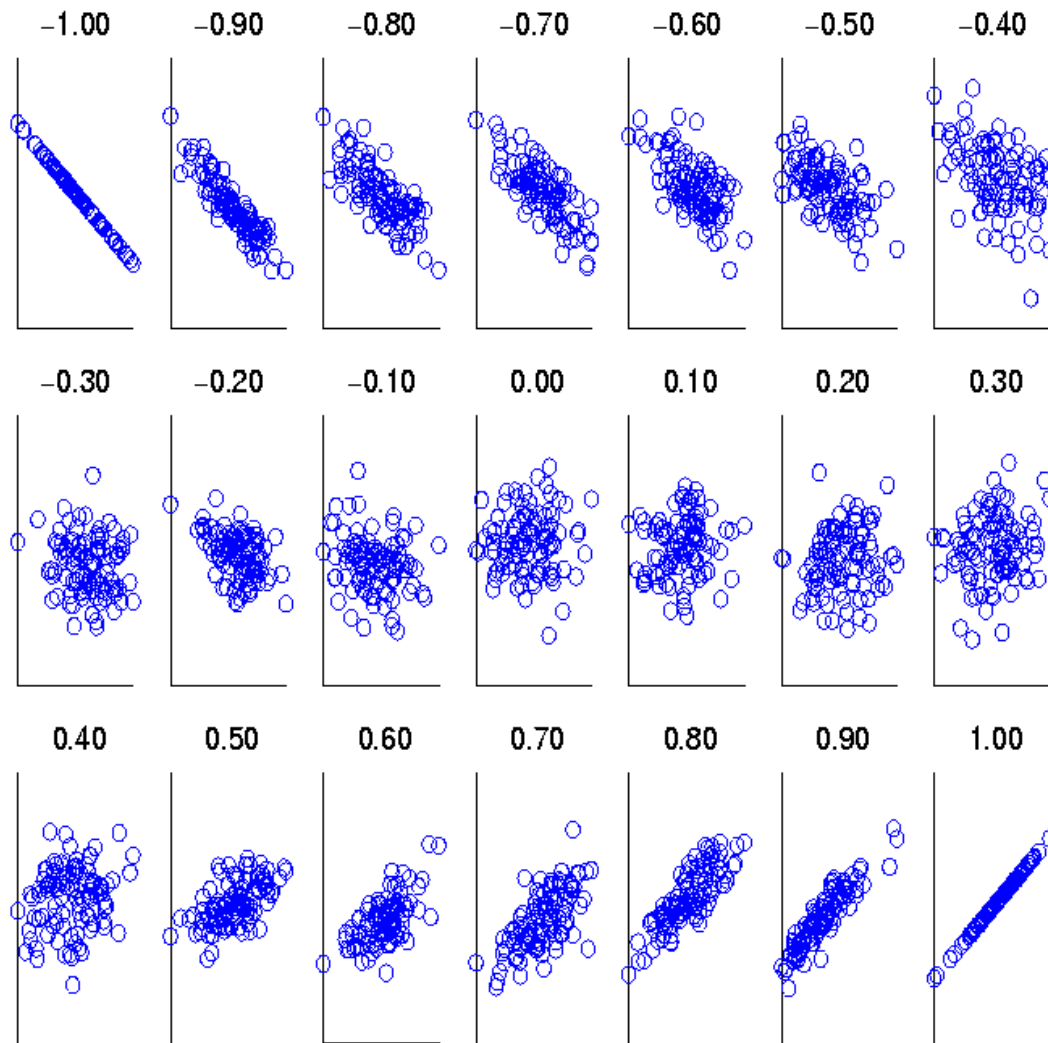
$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A\sigma_B}$$

- where n is the number of tuples
- $\bar{A}$ and $\bar{B}$ are the respective means of A and B ,
- σ_A and σ_B are the respective standard deviation of A and B ,
- $\sum(a_i b_i)$ is the sum of the AB cross-product.

Handling Redundancy: Correlation Analysis (Numeric Data)

- **Positive Correlation ($r > 0$):** As one variable increases, the other tends to increase.
 - **Example:** Height and weight are often positively correlated. Taller people tend to be heavier. The closer r is to 1, the stronger the positive relationship.
- **Negative Correlation ($r < 0$):** As one variable increases, the other tends to decrease.
 - **Example:** Temperature and hot chocolate sales might have negative covariance. Higher temperatures tend to mean fewer hot chocolate sales.
- **No Correlation ($r = 0$):** The variables don't seem to have any linear relationship.
 - **Example:** Shoe size and IQ are likely to have a correlation close to zero.

Visually Evaluating Correlation



- Scatter plots showing the similarity from -1 to 1 .
- The closer r is to -1 or 1 , the stronger the linear relationship.
- The closer r is to 0 , the weaker the linear relationship.

Correlation (viewed as linear relationship)

- Correlation measures the linear relationship between objects
- To compute correlation, we standardize data objects, A and B, and then take their dot product

$$a'_k = (a_k - \text{mean}(A)) / \text{std}(A)$$

$$b'_k = (b_k - \text{mean}(B)) / \text{std}(B)$$

$$\text{correlation}(A, B) = A' \bullet B'$$

Handling Redundancy: Covariance (Numeric Data)

- Covariance measures how two variables change together. It's similar to correlation, but instead of standardizing the data (making the standard deviation equal to one), **keep the original scales of the variables.**
- $$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

Correlation coefficient:
$$r_{A,B} = \frac{Cov(A, B)}{\sigma_A \sigma_B}$$

- where n is the number of tuples,
- $\bar{A}$ and $\bar{B}$ are the respective mean or **expected values** of A and B ,
- σ_A and σ_B are the respective standard deviation of A and B .

Handling Redundancy: Covariance (Numeric Data)

- **Positive covariance:** If $\text{CovA,B} > 0$ then A is greater than its average, B also tends to be greater than its average, and vice versa. They move up or down together.
 - **Example:** *Height and weight might have positive covariance. Taller people tend to be heavier.*
- **Negative covariance:** If $\text{CovA,B} < 0$ then A is greater than its average, B tends to be smaller than its average. They move in opposite directions.
 - **Example:** *Temperature and hot chocolate sales might have negative covariance. Higher temperatures tend to mean fewer hot chocolate sales.*
- **Independence:** $\text{CovA,B} = 0$ There's no linear relationship. Knowing the value of A doesn't tell you anything about B.
 - **Example:** *Shoe size and IQ score likely have zero (or very close to zero) covariance.*

Co-Variance: An Example

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

- It can be simplified in computation as

$$Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$

- Suppose two stocks A and B have the following prices in one week:
(2, 5), (3, 8), (5, 10), (4, 11), (6, 14).
- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?
 - $E(A) = (2 + 3 + 5 + 4 + 6) / 5 = 20/5 = 4$
 - $E(B) = (5 + 8 + 10 + 11 + 14) / 5 = 48/5 = 9.6$
 - $Cov(A, B) = (2 \times 5 + 3 \times 8 + 5 \times 10 + 4 \times 11 + 6 \times 14) / 5 - 4 \times 9.6 = 4$
- Thus, A and B rise together since $Cov(A, B) > 0$.

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Data Reduction

- **Data reduction:** Obtain a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- **Why data reduction?** A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set.

Data Reduction Strategies

- **Data reduction strategies**

- **Dimensionality reduction**, e.g., remove unimportant attributes
 - Wavelet transforms
 - Principal Components Analysis (PCA)
 - Feature subset selection, feature creation
- **Numerosity reduction** (some simply call it: Data Reduction)
 - Regression and Log-Linear Models
 - Histograms, clustering, sampling
 - Data cube aggregation
- **Data compression**

Data Reduction: Dimensionality Reduction

- **Curse of dimensionality**

- When dimensionality increases, data becomes increasingly sparse
- Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
- The possible combinations of subspaces will grow exponentially
- It becomes difficult to find meaningful patterns in the data.
- It requires more storage space and processing power.

- **Dimensionality reduction benefits:**

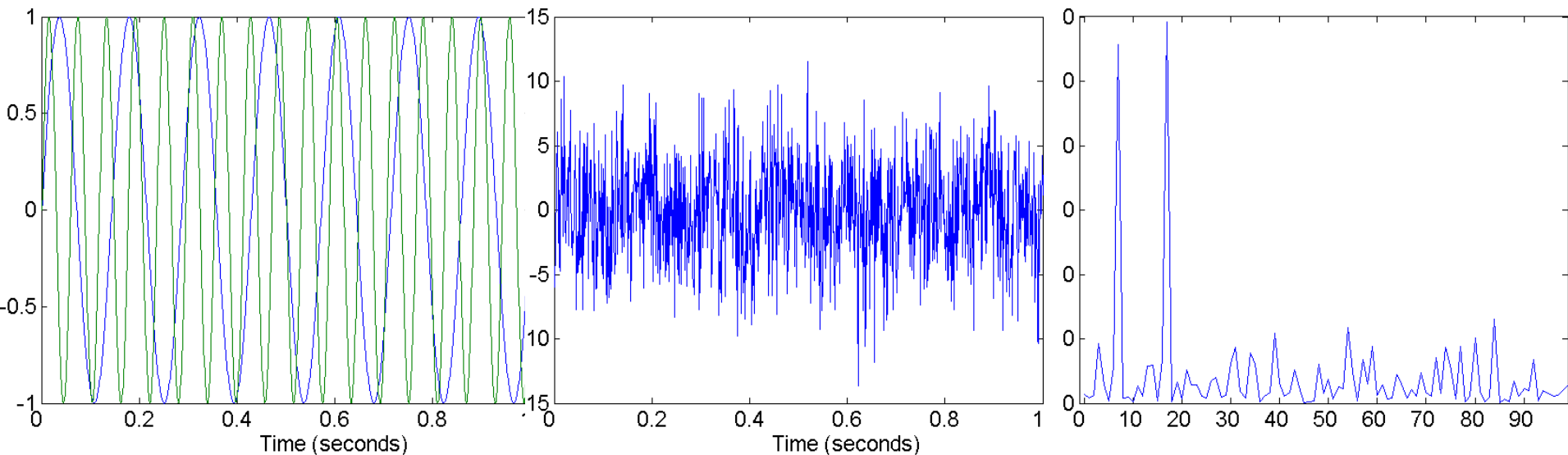
- Avoid the curse of dimensionality
- Help eliminate irrelevant features and reduce noise
- Reduce time and space required in data mining
- Allow easier visualization

Data Reduction: Dimensionality Reduction

- **Dimensionality reduction techniques:**
 - Wavelet transforms
 - Principal Component Analysis
 - Supervised and nonlinear techniques (e.g., feature selection)

Mapping Data to a New Space

■ Wavelet transform



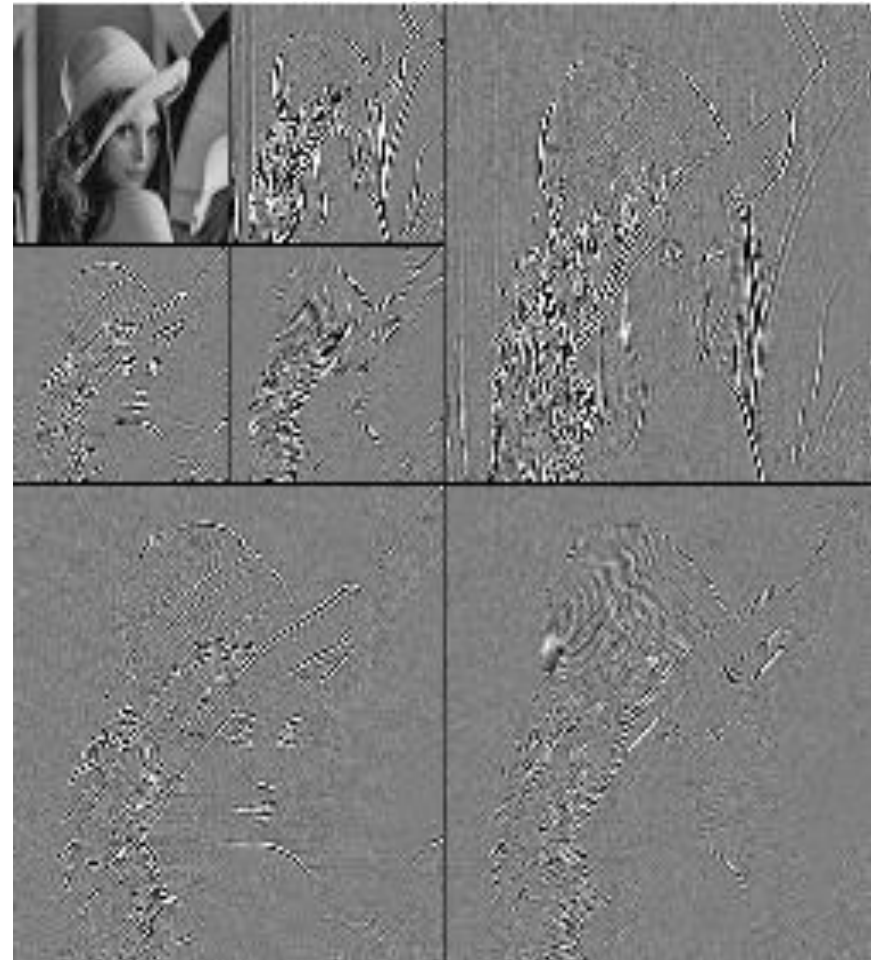
Two Sine Waves

Two Sine Waves + Noise

Frequency

What Is Wavelet Transform?

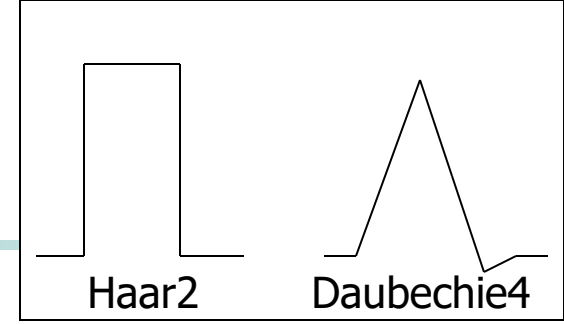
- These are mathematical techniques that can decompose data (signal) into different frequency components, to identify and discard less important components.
- it's powerful for analysing signals that have sudden changes or discontinuities. It breaks down a signal into "wavelets," which are localized waves of different scales.



What Is Wavelet Transform?

- The Wavelet Transform represents the signal in terms of these wavelets, specifying both the frequencies present and when (or where) they occur.
- Example:
 - **Image:** Wavelets are excellent for image compression (like JPEG 2000). They can identify sharp changes in an image (like edges) very efficiently.
 - **Audio:** If you have a recording with a sudden loud noise, wavelets can pinpoint exactly when that noise occurred.
 - **Medical:** Wavelets are used in analyzing EEG data, where sudden spikes or changes in the signal can be important.

Wavelet Transformation



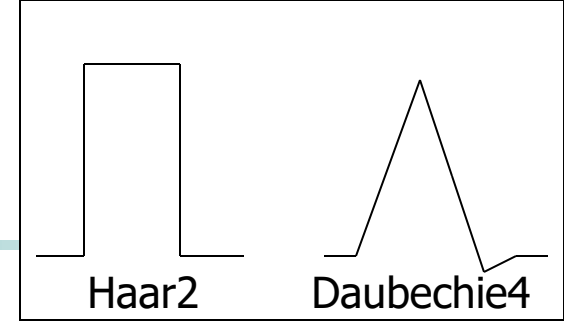
- **Discrete wavelet transform (DWT)** is a tool for analyzing signals at different levels of detail. It can focus on the most important parts of the signal and discard the rest, resulting in smaller file sizes without significantly sacrificing quality. **The DWT for:**
 - **linear signal processing, multi-resolution analysis:** The DWT is used for processing signals (like audio, images, or sensor data) and look at the signal at different levels of detail (multi-resolution).
 - **Compressed approximation:** The DWT stores only a small fraction of the strongest of the wavelet coefficients(focusing on the most important parts of the signal and discarding the less important parts).
 - **lossy compression:** some information is lost during compression, The DWT minimizes the impact on how the song sounds.

Wavelet Transformation

■ How the DWT Works

- **Padding:** The length of the data needs to be a power of 2. If it's not, add zeros to the end (padding).
- **Smoothing & Difference Functions:** The DWT uses two special functions: one that "**smooths**" the data (averaging nearby values) and one that calculates the "**difference**" between nearby values (like highlighting changes).
- **Recursive Application:** The DWT applies these two functions to pairs of data points. This process applies two functions recursively on the smoothed data until reaches the desired length (the desired level of detail).

Wavelet Transformation



- The Haar wavelet is a specific type of wavelet that is used in the Discrete Wavelet Transform (DWT).
- The Haar Wavelet Transform is a technique used to decompose a signal (like a sound or image) into different frequency components.

Wavelet Decomposition

- The table shows a simplified example of how the Haar works:

Resolution	Averages	Detail Coefficients
8	[2, 2, 0, 2, 3, 5, 4, 4]	[0, -1, -1, 0] [$\frac{1}{2}$, 0] [-1 $\frac{1}{4}$]
4	[2, 1, 4, 4]	
2	[1 $\frac{1}{2}$, 4]	
1	[2 $\frac{3}{4}$]	

- **Resolution:** This indicates the level of detail. Higher resolution (8) means more detail, lower resolution (1) means less detail (more general).
- **Averages:** These represent the smoothed or averaged values of the data at each resolution.
- **Detail Coefficients:** These capture the details or fluctuations that were lost when smoothing the data.

Wavelet Decomposition

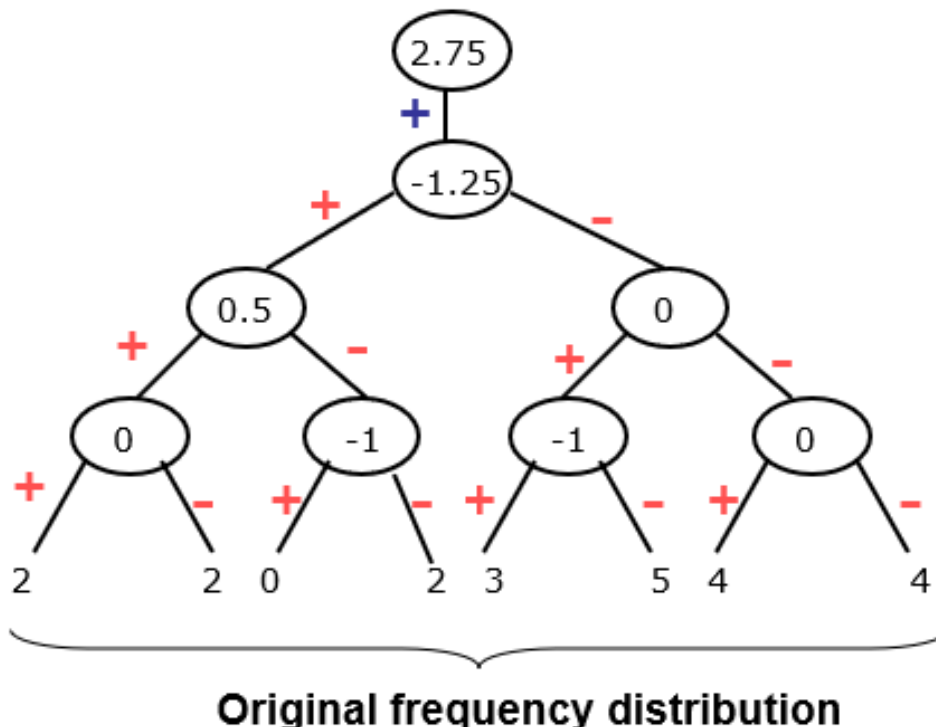
- **Starting Data (Resolution 8):** 8 data points (as measurements or samples of a signal (e.g., sound wave, temperature readings). e.g., sound wave, temperature readings). : **[2, 2, 0, 2, 3, 5, 4, 4]**.
- **First Transformation (Resolution 4):**
 - **Averages:** The DWT applies a "smoothing" function to pairs of data points and calculates the averages:
 - $(2+2)/2 = 2$
 - $(0+2)/2 = 1$
 - $(3+5)/2 = 4$
 - $(4+4)/2 = 4$
 - So the averages: **[2, 1, 4, 4]**.
 - **Detail Coefficients:** The DWT also calculates the "differences" or "details" lost in the smoothing process:
 - $(2-2)/2 = 0$
 - $(0-2)/2 = -1$
 - $(3-5)/2 = -1$
 - $(4-4)/2 = 0$
- **Second Transformation (Resolution 2):**

Resolution	Averages	Detail Coefficients
8	[2, 2, 0, 2, 3, 5, 4, 4]	
4	[2, 1, 4, 4]	[0, -1, -1, 0]
2	$[1\frac{1}{2}, 4]$	$[\frac{1}{2}, 0]$
1	$[2\frac{3}{4}]$	$[-1\frac{1}{4}]$

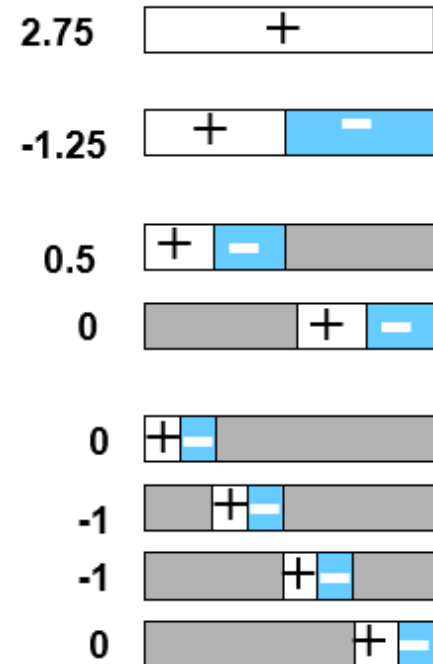
Wavelet Decomposition

- So $S = [2, 2, 0, 2, 3, 5, 4, 4]$ can be transformed to

$$\hat{S} = [2^{3/4}, -1^{1/4}, 1/2, 0, 0, -1, -1, 0]$$
- Drawing a tree for the Haar transform can be helpful to visualize the process of decomposing a signal into different frequency components.
- The Structure: Hierarchical Decomposition



Coefficient "Supports"



References

- Data Mining: Concepts and Techniques, Jiawei Han, Micheline Kamber and Jian Pei
- "Data Science for Business" by Foster Provost and Tom Fawcett